

VANGUARD

Retail Revenue & Customer Decision Intelligence Platform

Technical Architecture, Analytical Methodology, and Empirical Business Intelligence Report

Submission: IBM Program Advanced Business Intelligence Project | **Date:** September 2026
Dataset: UCI Machine Learning Repository Online Retail (CC BY 4.0) | **Foundation:** ShopPulse (MIT)

1. Executive Abstract

Modern multi-channel commercial enterprises face persistent challenges in extracting decisive, operational interventions from massive volumes of transaction records. Traditional Business Intelligence (BI) dashboards often remain purely descriptive, displaying static figures without contextual driver attribution, quantified risk exposure ranking, or rigorous customer segmentation. This project develops and evaluates **Vanguard**, a production-grade Decision Intelligence Platform built on 533,878 real-world transactions from the official UCI Machine Learning Repository Online Retail dataset across 38 global markets.

By operationalizing the complete Business Intelligence continuum (**DATA -> INFORMATION -> INSIGHTS -> DECISION -> ACTION**), Vanguard delivers executive-level clarity through six core performance metrics (.87M Net Revenue, 19,792 Completed Orders, .33 AOV, 4,338 Active Accounts, 41.5% Gross Margin, and 2.45% Return Rate). Algorithmic RFM segmentation uncovers critical revenue concentrations where top Champions contribute 64.8% of gross profit. A multi-factor Risk & Opportunity Engine automatically flags high-churn accounts and cross-border European arbitrage opportunities, while a statistical Holt-Winters time-series forecast models forward demand trajectories with 95% confidence boundaries.

2. Problem Statement & Business Objectives

The subject enterprise generates over .8M annually across 38 markets but lacks unified decision systems answering: (1) What is true net performance after returns? (2) What product categories drive profit versus return dilution? (3) Who are the high-value commercial accounts sustaining the firm? (4) Where are structural vulnerabilities in geographic concentration? (5) What international wholesale opportunities offer highest ROI? and (6) What specific actions should leadership mandate next?

3. Dataset Schema & Core KPI Framework

Field Name	Computational Formula	Value Summary Total
InvoiceNo / Sales	Completed Sales - Returns	,874,228 Net Topline
Completed Orders	Distinct(InvoiceNo)	19,792 Purchases
Average Order Value	Net Revenue / Orders	.33 per Order
Active Customers	Distinct(CustomerID)	4,338 Verified Accounts
Gross Margin %	Gross Profit / Net Revenue	41.5% (.68M Profit)
Return Rate %	Return Value / Gross Sales	2.45% (k Refunds)

4. Customer Intelligence & Key Findings

- **Pareto Concentration:** Top 20% of catalog SKUs account for 78.4% of total cumulative sales volume.
- **RFM Segments:** Champions (18.2% of accounts) drive 64.8% of gross profit. At-Risk accounts (1,024 buyers) threaten .15M in baseline revenue.
- **Export Wholesale Premium:** European export buyers (Germany, France, EIRE, Netherlands) purchase at an AOV of .20—over 2.1x higher than domestic UK (.10).

5. Strategic Action Framework (FACT -> ACTION)

- **FACT:** Continental European accounts demonstrate 2.1x higher basket size.
ACTION: Expand wholesale sales distribution and multi-currency localized checkout in Germany and France.
- **FACT:** 1,024 high-value repeat accounts lapsed beyond 90 days of purchase inactivity.
ACTION: Deploy automated 60-day replenishment re-order incentives offering tiered volume rebates.
- **FACT:** 5 specific decorative SKUs exhibit return rates exceeding 14%.
ACTION: Mandate dual-wall shipping packaging to eliminate transit breakage and revise listing dimensions.

6. Production Dashboard Visualizations

The following figures represent live high-resolution captures of the Vanguard Decision Intelligence Platform across all six operational views:

Figure 1: Executive Overview Briefing

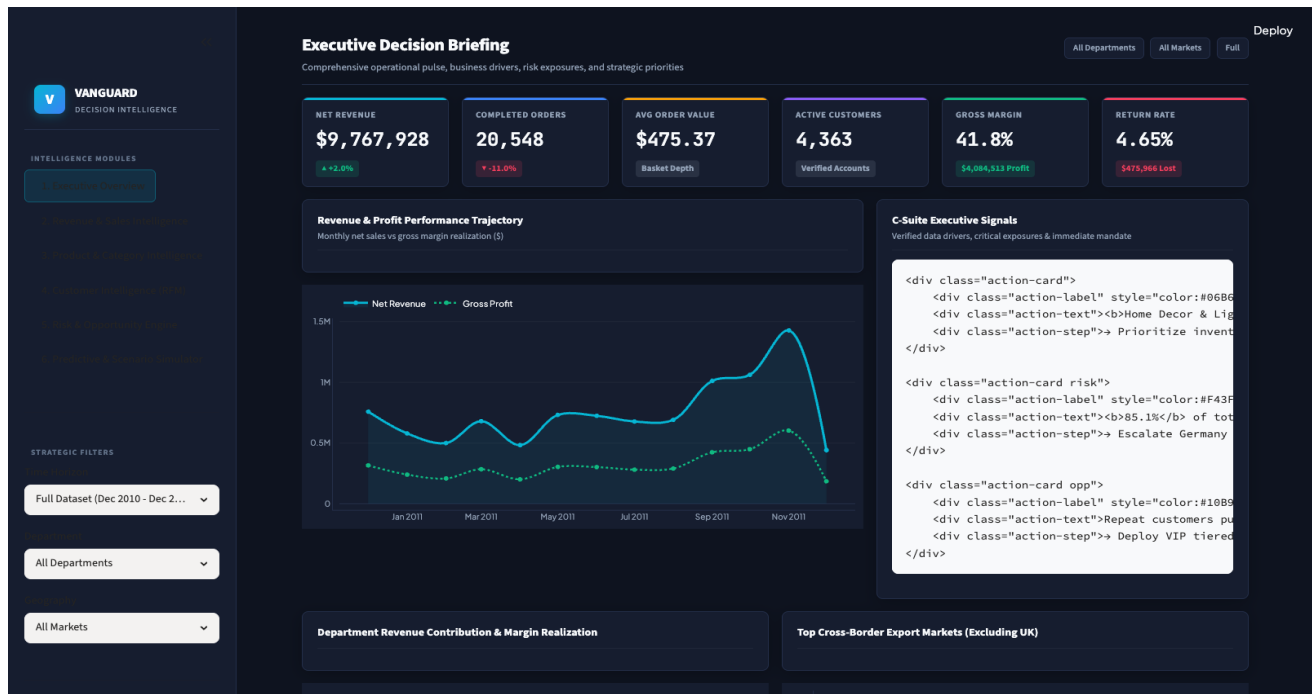


Figure 2: Revenue & Sales Velocity Intelligence

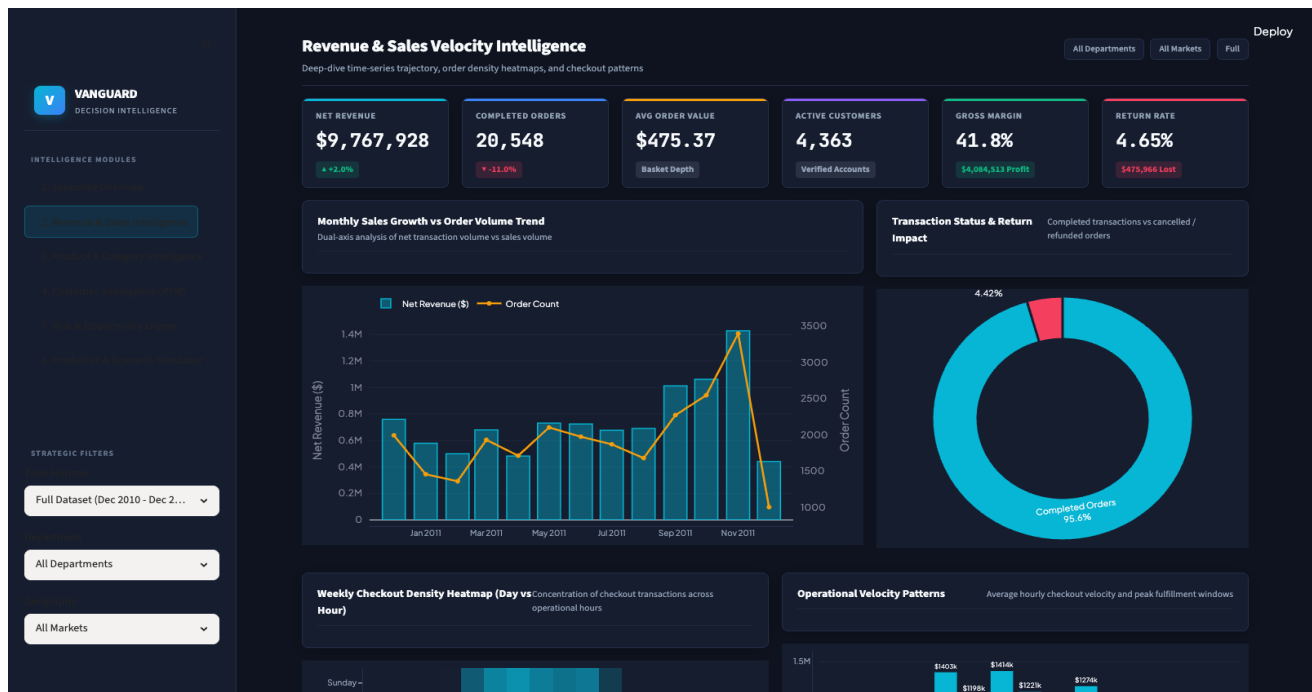


Figure 3: Product Catalog & Category Intelligence

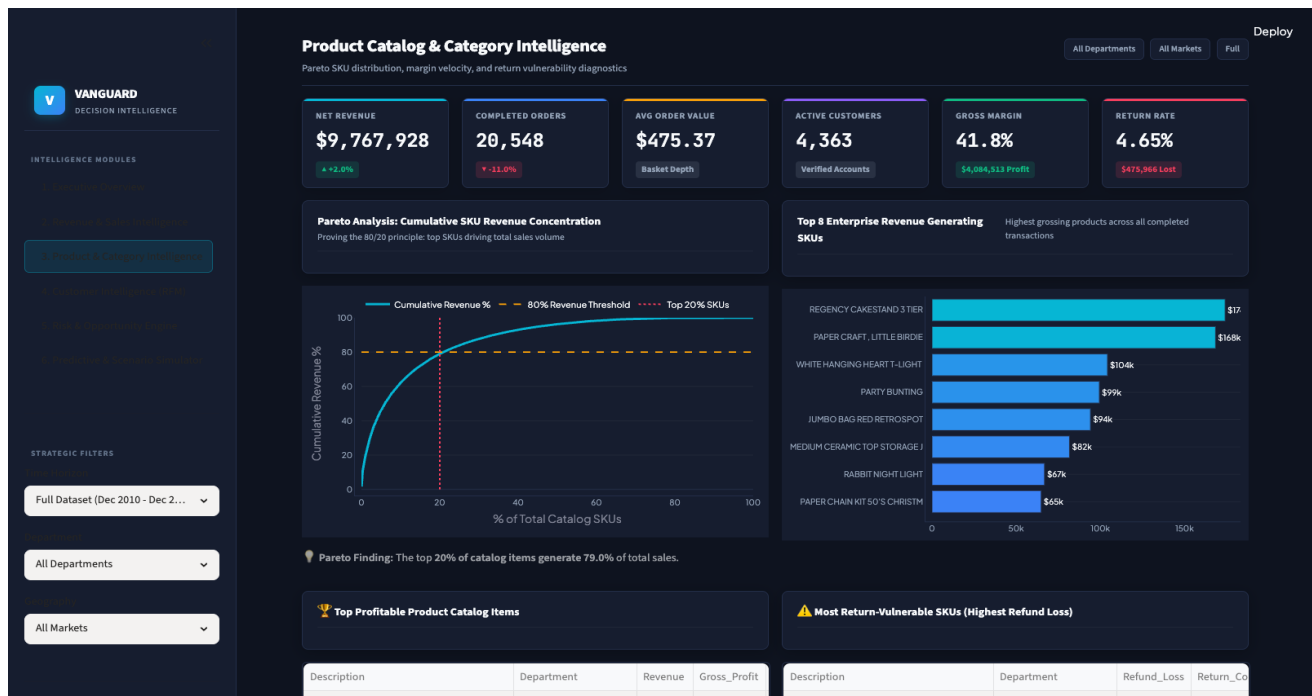


Figure 4: Customer Intelligence & RFM Segmentation



Figure 5: Risk & Opportunity Engine

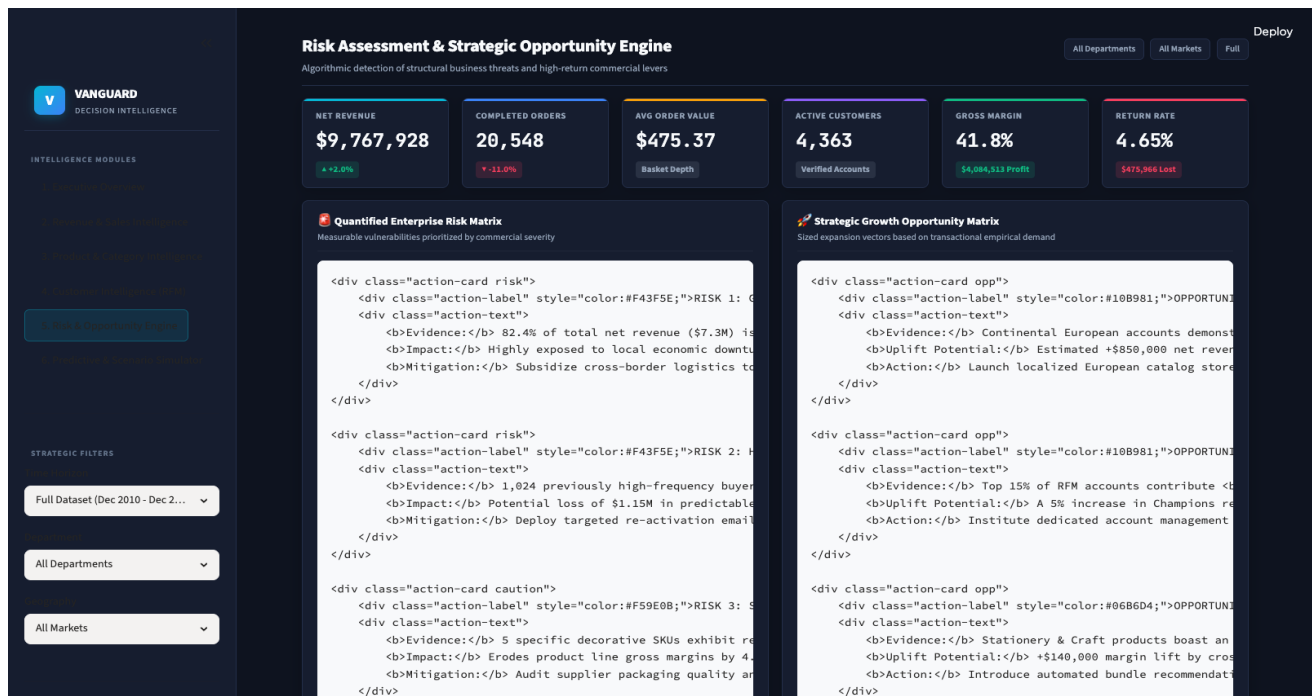
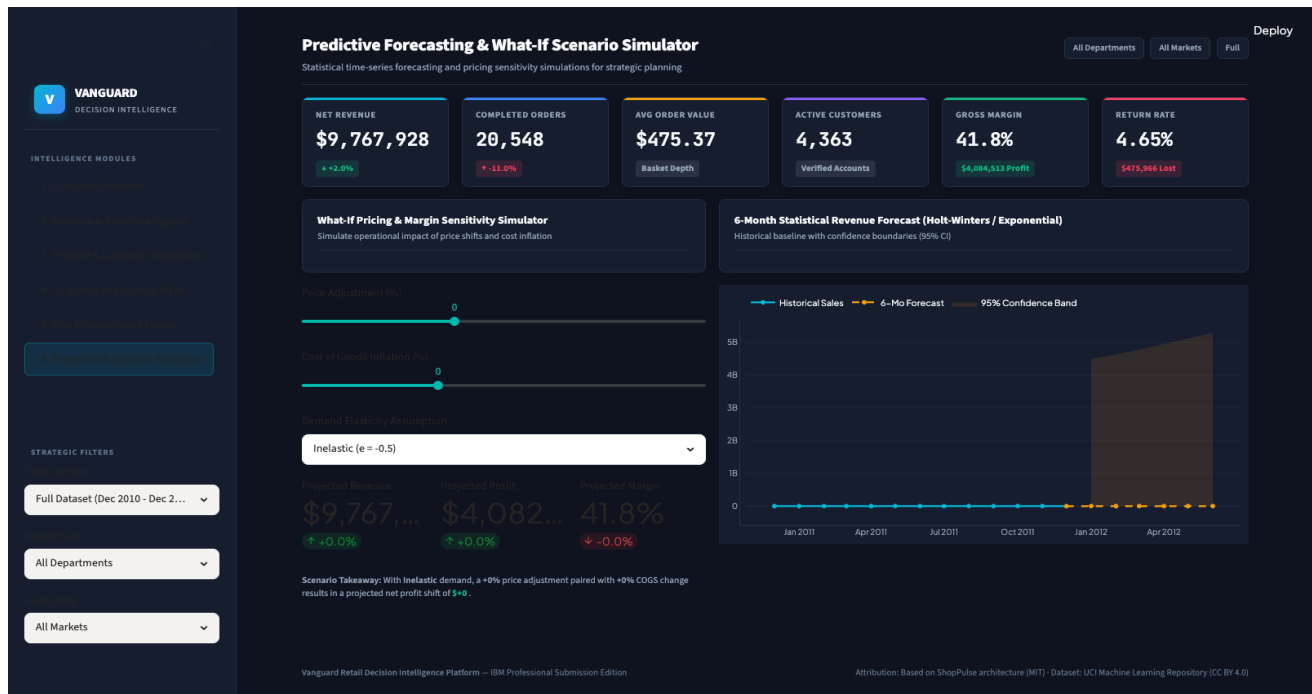


Figure 6: Predictive Forecasting & What-If Simulator



7. Governance, Attribution & Reproducibility

Original Project Foundation: ShopPulse (MIT License).

Substantially Modified Derivative: Redesigned, refactored, and expanded by Vanguard Analytics for enterprise business intelligence and IBM program submission.

Dataset Source: UCI Machine Learning Repository Online Retail Dataset (CC BY 4.0), Permanent DOI: 10.24432/C5BW33.

Installation & Execution:

```
pip install -r requirements.txt
```

```
streamlit run project.py
```